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1 COMP Neuromuscular junction

You should be able to: Trace transmission at the neuromuscular junction from the motor neuron action potential to the end plate potential and explain why every action potential in the motor neuron produces one in the muscle fiber.

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2 COMP Excitation contraction coupling

You should be able to: Trace excitation contraction coupling from the T tubule through the DHP and ryanodine receptors to calcium release from the sarcoplasmic reticulum.

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3

COMP

Crossbridge cycle

You should be able to: Order the steps of the crossbridge cycle, binding, power stroke, detachment, and reactivation, and state where ATP binds and where it is hydrolyzed.

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COMP

Calcium and the regulatory proteins

You should be able to: Explain how calcium binding to troponin moves tropomyosin off the myosin binding site and predict what happens to contraction if calcium cannot be released.

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5 Relaxation and calcium removal

COMP

You should be able to: Explain how SERCA and acetylcholinesterase end a contraction and predict the result if either one fails.

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6 The muscle twitch

COMP

You should be able to: Label the latent period, contraction phase, and relaxation phase on a twitch tracing and explain what limits each phase.

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7 COMP — Summation and tetanus

You should be able to: Explain wave summation and distinguish unfused from fused tetanus by stimulus frequency and calcium accumulation.

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8 COMP — Motor units and recruitment

You should be able to: Define the motor unit and explain how size of the unit and orderly recruitment grade the force a whole muscle produces.

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9 COMP — Length tension relationship

You should be able to: Interpret a length tension curve and explain why tension falls at very short and very long sarcomere lengths.

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10 COMP — Isometric and isotonic contraction

You should be able to: Distinguish isometric, concentric, and eccentric contractions and predict shortening velocity from the load applied.

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11 COMP — Muscle fiber types

You should be able to: Compare slow oxidative, fast oxidative glycolytic, and fast glycolytic fibers by speed, ATP source, fatigue resistance, and typical task.

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12 COMP — ATP sources during activity

You should be able to: Order creatine phosphate, anaerobic glycolysis, and oxidative phosphorylation by how quickly each supplies ATP and how long each can sustain it.

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13 COMP Fatigue and oxygen debt

You should be able to: Distinguish central from peripheral fatigue and explain the metabolic basis of excess post exercise oxygen consumption.

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14 COMP Muscle plasticity

You should be able to: Predict the change in fiber size, capillary supply, and mitochondrial content produced by endurance training, resistance training, and disuse.

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15 COMP Electromyography and grip fatigue

You should be able to: Record grip force and surface EMG over a sustained contraction and relate the decline in force to recruitment and fatigue.

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16 COMP Twitch and tetanus simulation

You should be able to: Generate twitch and tetanus tracings in a muscle simulation by varying stimulus voltage and frequency and identify threshold and maximal stimulus.

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17 COMP Smooth muscle contraction mechanism

You should be able to: Trace smooth muscle contraction through calcium entry and release to calmodulin and myosin light chain kinase and explain why the mechanism is slower and more economical than in skeletal muscle.

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18 COMP Smooth muscle types and regulation

You should be able to: Compare single unit and multi unit smooth muscle and list the stimuli that alter smooth muscle tone including stretch and local chemical signals.

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19

COMP

Cardiac muscle properties

You should be able to: Explain how intercalated discs and gap junctions allow the myocardium to act as a functional syncytium and state why cardiac muscle cannot be tetanized.

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COMP

Comparison of the three muscle types

You should be able to: Build a comparison of skeletal, cardiac, and smooth muscle by calcium source, regulatory protein, control, and speed of contraction.

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